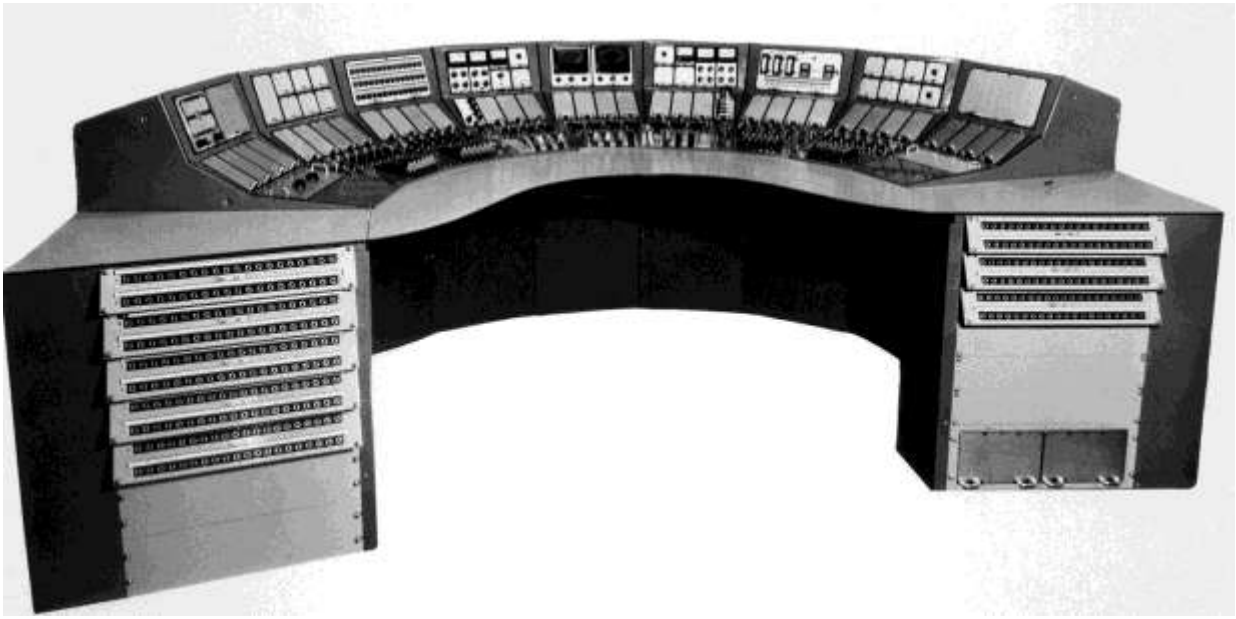
1960's – PYE BROADCAST AUDIO CONSOLES | PART FOUR | THE INNOVATIVE PYE TRANSISTOR TV STUDIO MIXER

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THE FIRST MODULAR BROADCAST STUDIO MIXER



One of the Pye consoles supplied to ATV Elstree, still with many blanking panels fitted.
Photo: Pye via Richard Ellis

The introduction of transistors at the end of the '50's allowed audio desks to be smaller….or packed with more goodies. In 1960 the Pye audio department under Bob Barrass, designed a big germanium transistor desk which was first mixer design that I've seen that was built in a fully 'modular' style with plug in 'cassettes', as they were named by Pye, located within the desk. [1] The 'Auxiliary' routing modules, along with a basic 'Mic Amp', or a version of the 'Mic Amp complete with Equalisation' were put into the desk immediately adjacent to the faders.
This innovative console was definitely the best studio mixing console design in those very early years of transistor designs, and it also was given a most original layout feature, a curving 'wrap-around' layout. This brought all the faders and other controls within easy reach of the person mixing, whilst still maintaining him in the 'sweet spot' for the monitoring loudspeaker.
This idea of a semi-circular console layout was just so obvious that it's surprising that the next time it appeared was when Dick Swettenham and Keith Grant designed their first console for Olympic Studios some years later. Their company Helios then used the semi-circular layout often. The BBC also stayed with the same design idea for many years for consoles that were either built by them, like the TV Type 'D' desks or purchased for them, by getting both Neve and Calrec to follow the this layout for both BBC Radio and TV desks.

ATV's four 34-channel consoles for Elstree

The first company that bought his big new Pye design in mixers was Associated Television at Elstree, starting in 1960.

In 1955 ATV had gained the ITV contract for 'Weekends' in London and also the contract for 'Weekdays' in the Midlands. ATV's boss, Lew Grade then took over the old Elstree Film Studios in 1958 and started converting them for television. The first two studios, which were extensively modified from existing film stages were Studio 'D', which opened in November 1960 and Studio 'C' which opened in January 1961. These were fitted with the new Pye 34-channel consoles, to be followed by two more as Studio 'A' and 'B' were being constructed from new and they likewise had the identical consoles of 34-channels.
In late 1961 Dave Langridge had just been taken on in the ATV sound department after working at EMI Abbey Road and remembered being shown around his new workplace at Elstree: "*When I was taken to A or B to see one of the new studios, definitely the new desk was there, because they were plugging in modules and testing equipment.*" [2] It wasn't until September 1961 that the newly built Studio A came into service, followed by the final studio, B, which opened just before the Christmas 1961.[3]

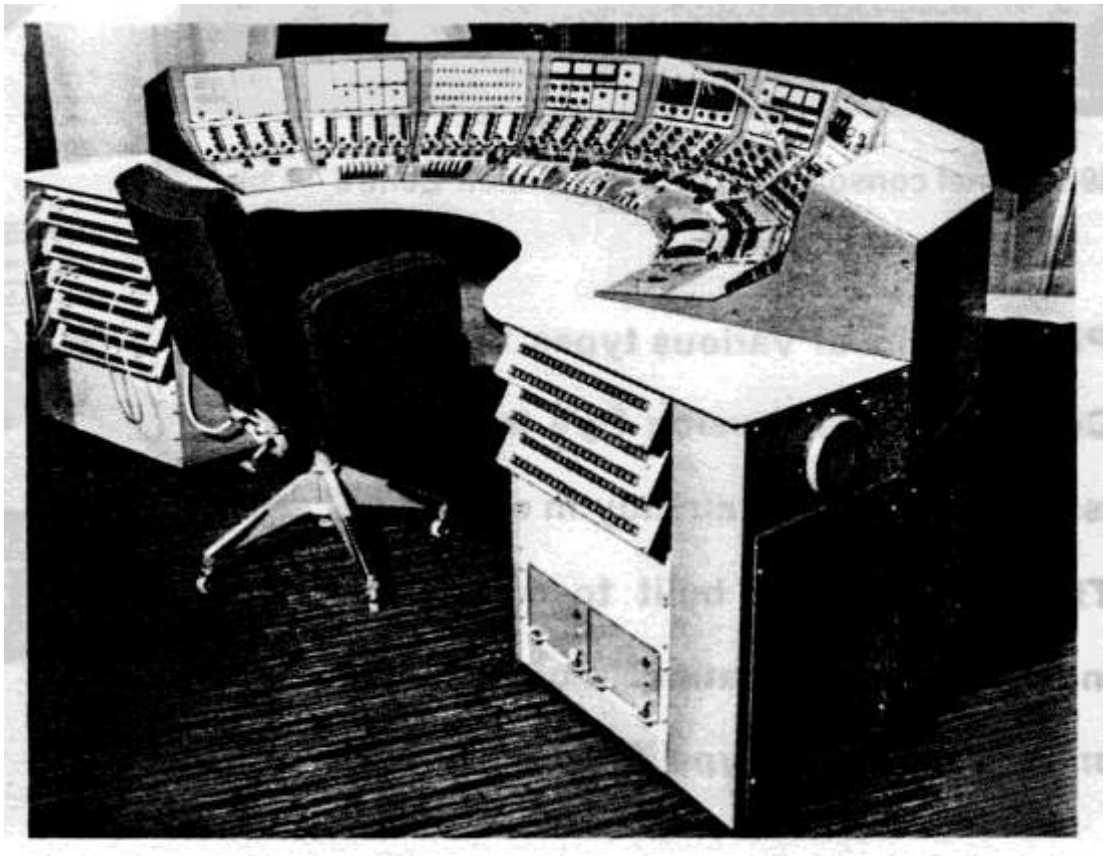


One of the four identical 34-channel consoles that went to ATV Elstree.
Photo: via Brian Summers

The console in both of the two photos above is obviously still at Pye awaiting delivery, and it has lots of missing modules covered with face plates, but we can see how ergonomic that 'wrap-around' design was for the sound mixer, putting all the controls around him. No need for a 'chair on rails' as often seemed the case once the even bigger Neve's arrived in later years. The new transistor design, having allowed all the Mic Amps to come out of the bays and into the console 'work surface' also made possible a great degree of customisation around the mixing position.
The newer 'thin' Painton Quadrant faders had been introduced in mid 1959 and these allowed the faders to be closer together for easier control when using many faders at once; this was great for say a 'chat' show with multiple mics to being handled or a light entertainment show where you were juggling a band, singer and audience at the same time.
The only criticism I guess is that these 'handy' groupings of the thin faders meant that the fader didn't actually line-up totally with their associated channel module. These 'cassettes' in the Pye desks were also still too wide to all be fitted into the one wedge section really close to their associated faders anyway, as we will see – but a new design concept had been born.

Here's a photo of one of the ATV 34-channel desks 'fully fitted out'. Immediately above the groupings of six of those Painton thin faders, are the 'Mic Amp with Eq' modules. Their size though only allowed four to be fitted in each 'wedge'. So the remaining two faders would just have been

working with ‘Mic Amps’ alone. Hopefully these layout details will become clearer in awhile when we look at some BBC documentation.



One of the ATV desks with the Mic amp/EQ modules fitted.
Photo Pye brochure via Brian Summers

Looking at these consoles, you realise the degree of customising being undertaken by Pye. There are differing numbers of the ‘bays or wedges’ depending on the total number of channels fitted, and the modules within each are either of the plug-in ‘cassette’ design that the Pye audio team under Bob Barrass had developed, or fitted with custom made panels to suit each customers differing requirements.

All the desks, like the Elstree ones which had nine bays/wedges, have faders as normally fitted, flat on the console, with the next section directly in front of them being very slightly angled. This housed the ‘Auxiliary’ module unit with the Group selector switch, a PA switch and Echo and Foldback sends. Next up came another more upward angled section with the ‘Mic Amp’ modules; which as I’ve mentioned were usually four of the ‘Mic Amp with Eq’ and two with just the standard ‘Mic amps’, although that could be altered as required.
Finally the top, almost vertical section had more straight Mic/Line Amps and some of the Pye Comp/Limiters. This section is also where the special panels could be located, with the Talkback keys, or Talkback Speakers, ‘Effects’ switching units or remote controls for the EMT140 Echo plates.
As both the ‘Mic Amp’ or ‘Mic Amp/Eq’ cassettes were interchangeable, it’s obvious that a common ‘back connector’ must have been in use.
The ATV desks are fitted with four of the Pye 4060 Compressor/Limiters and I’ll take a look at those units later in this article.

A look at the modular ‘cassettes’

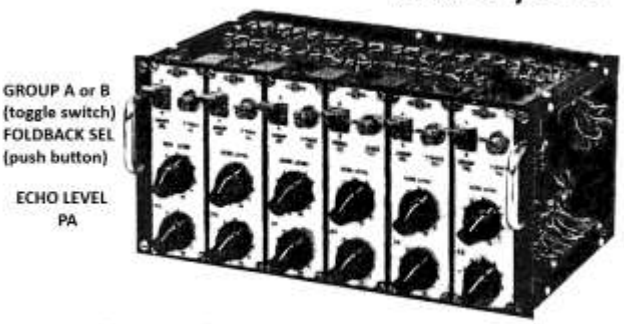
Here’s some Pye brochure images to explain some of the module ‘cassettes’:

Mic Amp with Eq



This unit is intended to function as a microphone channel amplifier, but its characteristics are such that it may also be used as a high level amplifier with inputs up to —20 dBm. The amplifier, which is of advanced design, is provided with high and low frequency compensation with a constant slope variable turnover characteristic, and a mid-lift or “ Presence ” control with a selection of four frequencies.

Auxiliary Unit



This is an auxiliary unit containing miscellaneous channel controls including Foldback and Public Address selection, Echo Ratio Control and Group Selection.

Compressor/Limiter

Equivalent in proportions to two of the basic minimum size cassettes, this unit may be switched to operate as either a limiting or a compression amplifier. The compression ratio is adjustable over a very wide range.



The Mic Amp/EQ Unit above differs from the ones we’ll see later in the BBC desks in having different control knobs and switches. The ‘Auxiliary Unit’ has at the top a toggle switch for the pair of groups it could be assigned to, along with Foldback and Echo pots and a PA on/off switch. Echo in the BBC desks was in fact an ‘echo mixture’ control that is described later and even the Compressor/Limiter differs a bit in the BBC Desks.



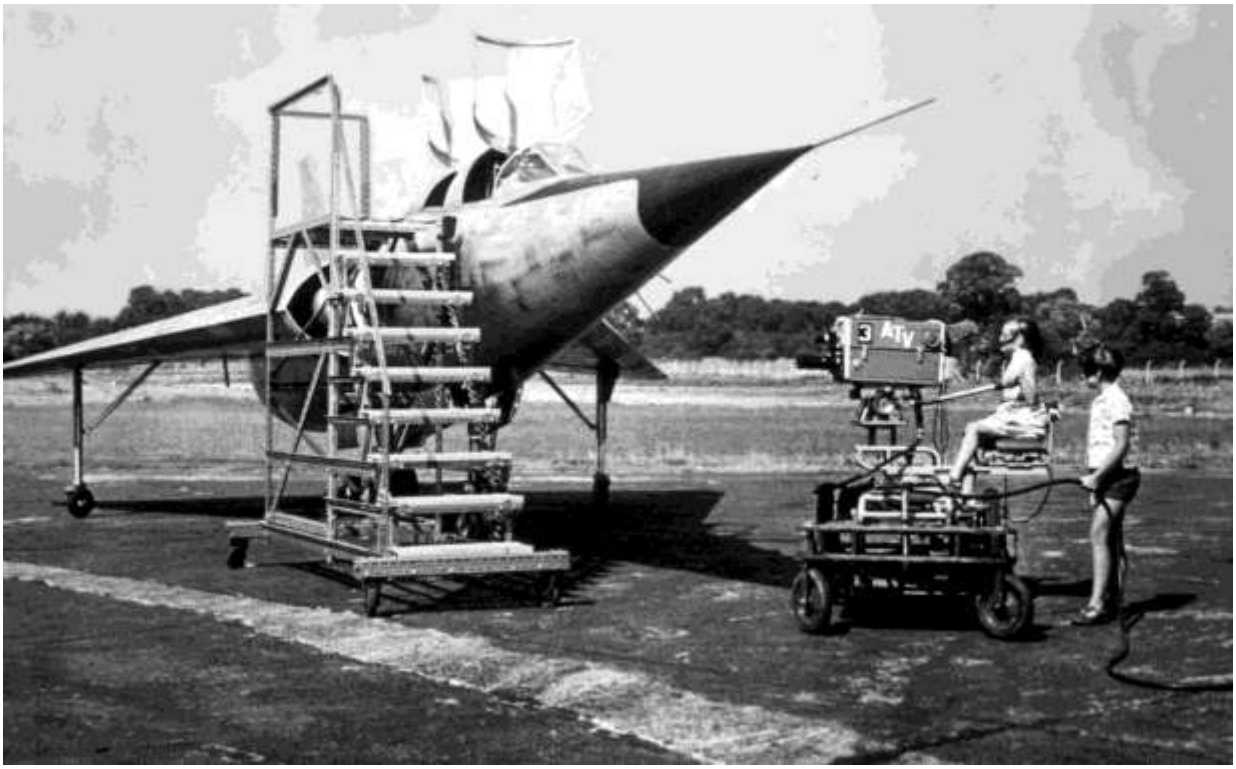
This is the Studio B Sound Control Room at Elstree showing the Pye TVT console and 2 Studer J37 tape decks along with an EMT 930 turntable. I love the reading material displayed on the grams script rack!
Photo: EMT brochure

Studio 'B' at Elstree with its superb grams equipment. The control rooms at Elstree were built across a corner of the studio and like all control rooms in those days, was looking down onto the studio floor.

Enough 'Drama' for the Gram Op

Like all the major studios, ATV Elstree made a very wide range of programmes in its 4 studios of which Studio A was 6400 sq ft, with Studio B a little bit bigger at 6720 sq ft, and the newer Studio's C and D both being the biggest at 9200 sq ft. Many of the Elstree programmes were for sale to the US, mainly Light Entertainment shows in the big Studios C and D I think, but among its other UK shows were studio dramas, which had huge audiences at that time.

Some of the ATV dramas were long running like Emergency Ward 10, but I thought I'd find a section of a studio drama to explain some of production and sound techniques of the day.



Surely ATV's youngest camera crew manning a Pye Mk111 camera in front of the Scott Furlong Predator, a model 'delta winged jet' built for the ATV series 'The Plane Makers'.
Photo: via Richard Ellis

TV studio drama's got large viewing audiences and one of ATV's was 'The Plane Makers', about which Wikipedia tells us:
"The Plane Makers is a British television series created by Wilfred Greatorex and produced by Rex Firkin. ATV made three series for ITV between 1963 and 1965. It was succeeded by The Power Game, which ran for an additional three series from 1965 to 1969."
A TV Times article reprinted on the Transdiffusion website gives an insight into the preparation of a show:
"During the first week of rehearsals, the No. 1 cameraman and the lighting director will work out approximate camera angles and lighting for the sets, and the wardrobe department will take measurements.
Sound mixer Pete Lodge also will have started work on his sound effects and background noises — all those whirring machines we see on the screen aren't making any noise at all. What we hear is recorded at Handley Page. And outside broadcast planner Ross Compton will have "taped" the exterior shots. For although the programme is live, about 10 per cent of each show is actually exteriors.
At 8.30 a.m. on the day before the show, camera rehearsals start. The sets go up and the electricians start rigging the lights. By 4.30 p.m., everything is ready for the rehearsals which may go on until late in the evening. On transmission day rehearsals begin in the morning and run straight into the afternoon dress rehearsals. At 7.30 p.m. final checks are made on the technical side, and at 8 p.m. the show goes on." **[4]**

‘The Plane Makers’ being recorded in September 1963

Imagine we are in the sound control of Elstree Studio B (as in the photo above), it’s near the end of the show which we’ve been recording since 8pm, so I guess it’s about 9.30 now. It’s hours since we rehearsed this scene and we’ve had lots to think about since, so neither the Sound Director, as he was called in ATV or the Gram Op, recall that much apart from what they had scribbled alongside the dialogue on the scripts in front of them.
In this clip we see four separate scenes are involved, plus some telecine film playbacks:
1: OB Insert -The aircraft factory/airfield exterior. This has been shot on video, which entailed the use of large TV cameras using an OB scanner plus recording to 2″ videotape, all of which made it ‘very cumbersome’ at times. It was still preferred by the Producers though because it matched the quality of the video studio scenes far better than did any filmed exterior scenes. The plane is a full size model, that can probably move on it’s own, but nothing more. A necessary ‘star’ though in this series.
2: Studio set 1 – The cockpit interior of the new jet airliner, about to undertake it’s first flight. There was underlying tension to this because the flight has been brought forward without the usual amount of testing, to beat that of a rival plane. Do note that the airfield outside the cockpit side windows is reproduced from film using ‘back-projection’. However that film has to start to move, just as the plane does, so timing it is critical.
3: Studio set 2 – The Control Tower, where the tycoon, played by Patrick Wymark with his backers are watching the take-off.
4: Studio set 3 – Sitting room where an old lady is anxious about the event (sorry I haven’t delved into the story-line to know her concerns).
Plus some more Telecine Film – Apart from the back-projector running behind the cockpit scenes, we also have a short film insert of the pilot’s view during take-off and a model shot of the aircraft lifting off and in flight…. a rather tacky bit of film that…but hey it’s a TV budget here…and 1963!

Now if we think of this from the sound mixer’s point of view, he has booms in the 3 studio sets to adjust, plus any sound on those ‘inserts’ and the gram op has both jet engine effects and ‘radio speech’ to vary in level across the vision cuts. There’s also some dramatic music to play in.
Dave Langridge remembered that when there were scenes based in the aircraft factory, that the Gram Op played the background factory FX over slung foldback speakers to remind the actors to keep their voices up!
Don’t forget, in 1963 when this was made videotape had only been around in British TV since 1958 and ‘edits’ were still just that; physically cutting the 2″ tape which was a ‘surgical operation’ requiring a microscope and a dose of ‘magic solution’, that identified the control track pulses which needed to remain continuous. [6] TV drama’s like this Plane Makers therefore avoided cutting the tape and productions were made in sequences usually the length of ‘each part’- this being commercial TV of course.
It’s nothing special really, only a ‘typical episode’ …..but let’s take a look anyway:
TO VIEW THIS VIDEO PLEASE SEE THE ORIGINAL ARTICLE AT:
www.postfade.co.uk

'The Plane Makers' – A short excerpt from 'Too Much To Lose' Series 2 Episode 1

I don't think the Gram Op would be playing 78rpm FX discs for the 'jet noise', and because of the involvement of Handley Page at the nearby Radlett airfield in quite a lot of the programmes, I guess he's been out and recorded some jet engine sound FX on an EMI L2 portable battery tape machine. These would now be running on possibly two of the 1/4″ J37 tape-decks we've seen in the photo of the Elstree 'B' sound control room, but it's fairly likely that he'd have them dubbed to separate tracks on one deck, and be switching tracks with his 'grams faders' as required. This lets him 'start engines' and then accelerate etc.
Then there's the 'RT' from the plane. It switches to 'distort' when heard in the Tower and on the airfield shots. I'll explain the 'telephone distort' unit built into the Pye desks later, but it's not used in this and it's most likely they would have some pre-recorded 'distorted RT' in this case, on another tape deck. He has some dramatic music cued up on the EMT turntable as well.
Everyone has their Camera Scripts, carefully labelled with their own relevant instructions and the big question then becomes…in a live continuous action sequence…when do we actually 'cut' between these scenes? Realising that just 'following the pictures', will inevitably mean you're going to be a fraction late, good Director will lead you in, by saying "Coming to Camera 3…and 'Cut'!" At that point the Gram Op 'chops' his background jet background or possibly switches to another track. He obviously has to start and stop the tape decks as required and re-cue them as well!
The Sound Mixer, meanwhile does a very rapid cross fade between the booms covering the studio dialogue and you in fact hear him do that during a word going across from the cockpit to the old lady at one point; he just has to 'go with the cut'. There's the OB Sound to fade as well, but there were sometimes parts of the show when the sound mixer did have the easier job compared with the Gram Op or Boom Ops…as all Sound Supervisors will own up to.
Also note that the Lighting Director here is trying to impart some movement in the cockpit by varying his lights and the Director had the OB cameraman do a couple of short 'tracks' to likewise help the feeling of movement of that 'big model plane'.

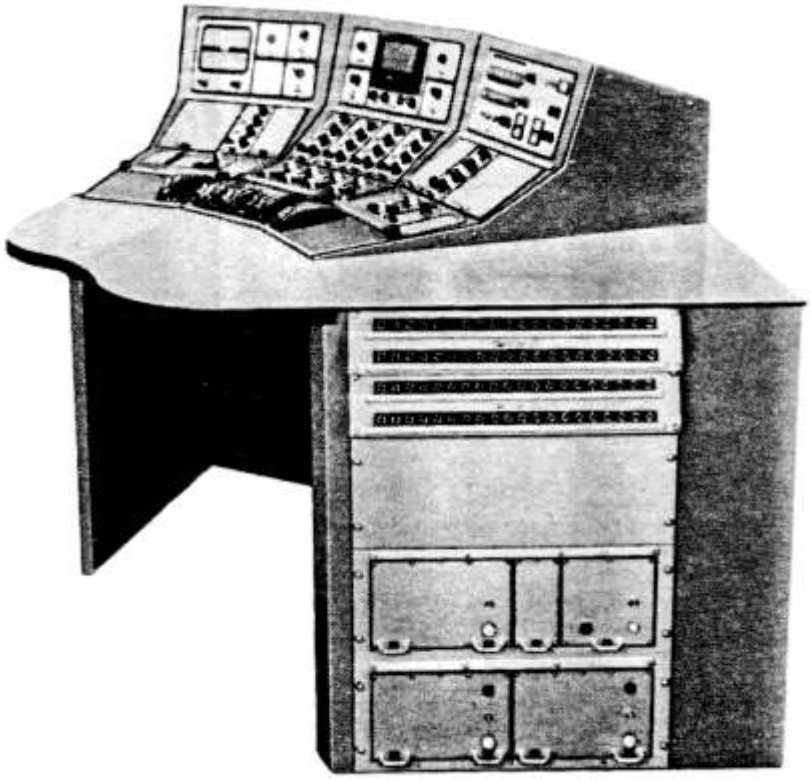
Memories of the last of Elstree's Pye's

ATV, having been the first to install the Pye desks, were also the first to replace them with the new Neve's. Richard Knight recalled: ***"I joined ATV Elstree in 1973, Studio C and D had switched to Neves but Studio B still had the Pye. Studio A's control room had been stripped and the studio floor was run from B's control room. As a trainee our job was to do the daily alignment on the desks. As I remember it:***
Remove one of the two PPMs and replace with the VU meter.
Adjust the VU for mechanical zero.
Put the desk in line-up and adjust the tone oscillator level to read -4db (I think) on the VU.
Adjust the mechanical zero of the PPM, as I remember that was to the right.
Put all the meters back to their normal position and adjust the PPMs for electrical zero, needle to the left.

Put the desk in line-up again and adjust the PPMs to read 4.
Join the rest of the crew for breakfast!" **[7]**

Some Pye's for other clients

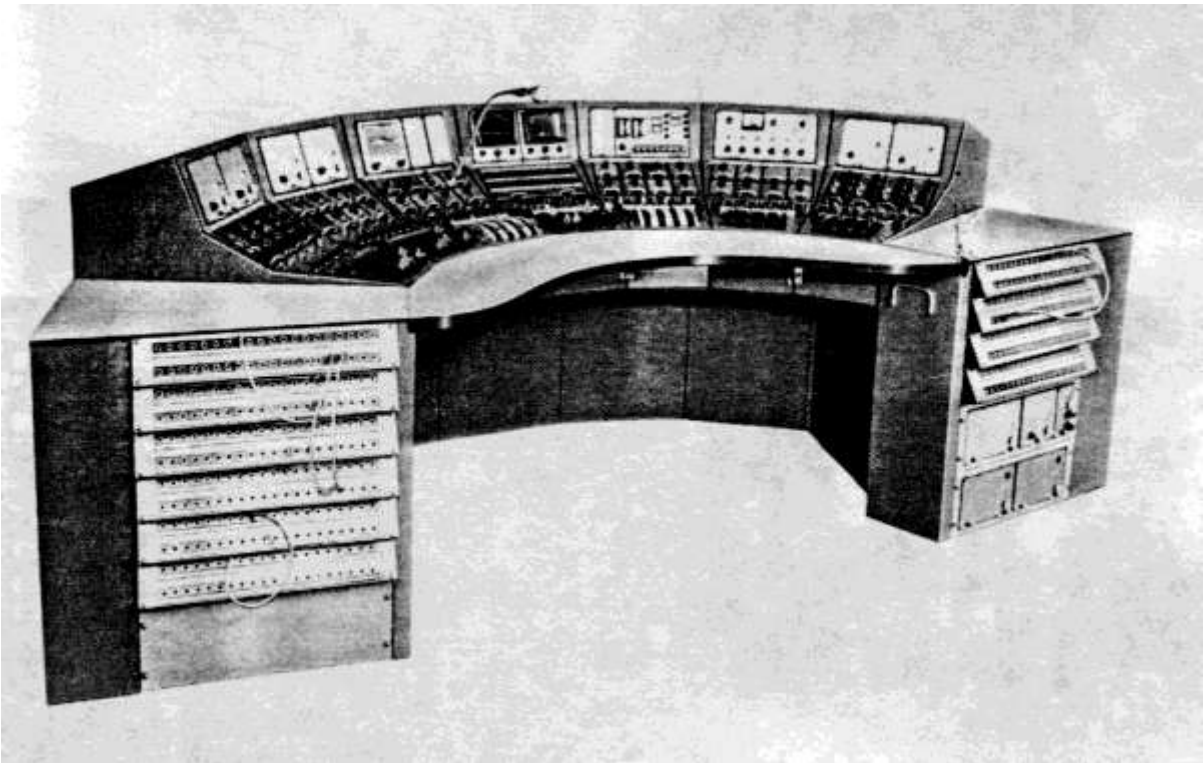
We'll see some more details of the cassettes a little further on, but let's look at a few illustrations from the Pye brochure:



8-channel desk for TWW Wales

The desks didn't have to be 'big' of course, but they still curved.

This is a small 8-channel version for Television Wales and the West, who had part of the split Welsh ITV franchise in its early years. There are eight faders, six centrally with four of the 'Mic Amp/EQ' cassettes, plus another on either side, also with the Eq.



A 24 channel model for Tyne Tees.

A 24-channel desk for ITV's Tyne Tees and the layout has two PPM's but only one 4060 Comp/Lim, plus 2 EMT140 Remotes.

The BBC equip with even bigger desks

The new BBC Television Centre came into operation in phases and it officially opened with TC3 operational on 29th June 1960. TC2, 4 and 5 opened over the following 14 months in that order. [8]
After those first studios were built, the construction of 3 more, requiring more facilities got underway.
A BBC produced booklet, issued after the opening in 1960 stated:
"Four of the seven production studios are now in use. Studio 3, for general purposes, and Studio 4, for light entertainment, music and drama, are each 100 ft x 80 ft x 40 ft high (30 m X 23 m x 13.5 m).
Studio 2, for general purposes, and Studio S, specially designed and serviced for schools programmes, are each 70 feet x 50 feet x 33 feet high (21 m x 15 m x 10 m).
***The other three studios will be brought into operation as they become equipped and of these, Studio I will be the biggest in the Centre, measuring 108 ft x 100 ft x 54 ft high (33 m x 30 m x 16.5 m). It is intended for light entertainment and music programmes and will be able to accommodate an audience of six hundred. Studio 6, for general purposes, will be the same size as Studios 3 and 4. Studio 7, for talks and discussion programmes, will be the same size as Studios 2 and 5. In addition to these studios in the Centre itself, there are three others in nearby Lime Grove, as well as the BBC's Television Theatre at Shepherds Bush."*[9]**
The BBC had always constructed their own sound desks and certainly the first studios at TVC had fairly basic BBC Type 'B' consoles like this:

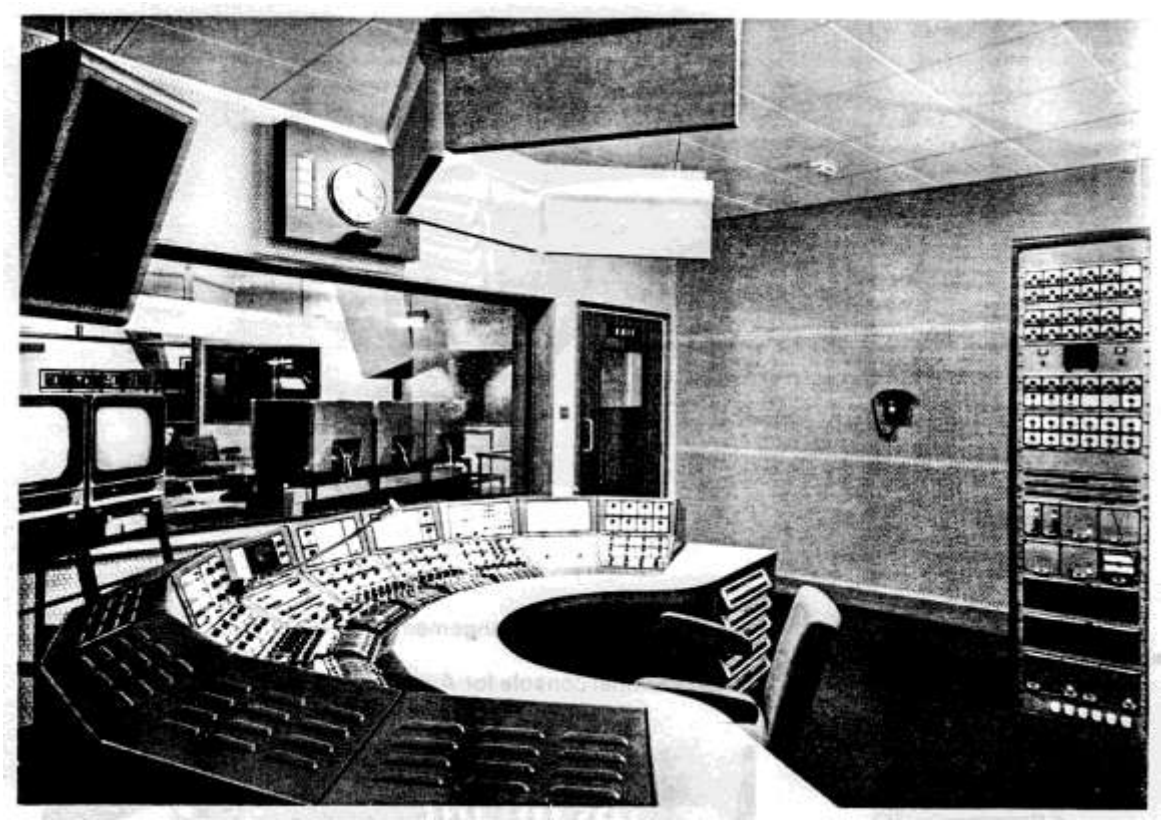


Brian Hiles mixing 'Adam Adamant' on a BBC Type 'B' desk in TC4, 1966 or '67. The caption to this photo says: " Note 'The' TV Centre limiter in left of frame (GB Kalee). It was the only one and was bookable on a first come, first served basis."
Photo: Bob Foley via Tech Ops website

Note the old style 'big' Painton faders on that BBC desk. It must have been a thrill for the Pye audio department to be awarded a BBC contract that eventually came to five of their new desks, three at TVC and one at the TV Theatre and even one for BBC Radio…..and the BBC staff must have been pleased too, compared with those primitive looking Type 'B' desks already in TVC.

The Biggest UK broadcast desk

The BBC's biggest studio, TC1 was planned to be opened in the summer of 1963, but was delayed and it opened on 15th April 1964 and it had a 43-channel Pye desk, which was a large sound console in the early '60's and was the biggest Pye built:

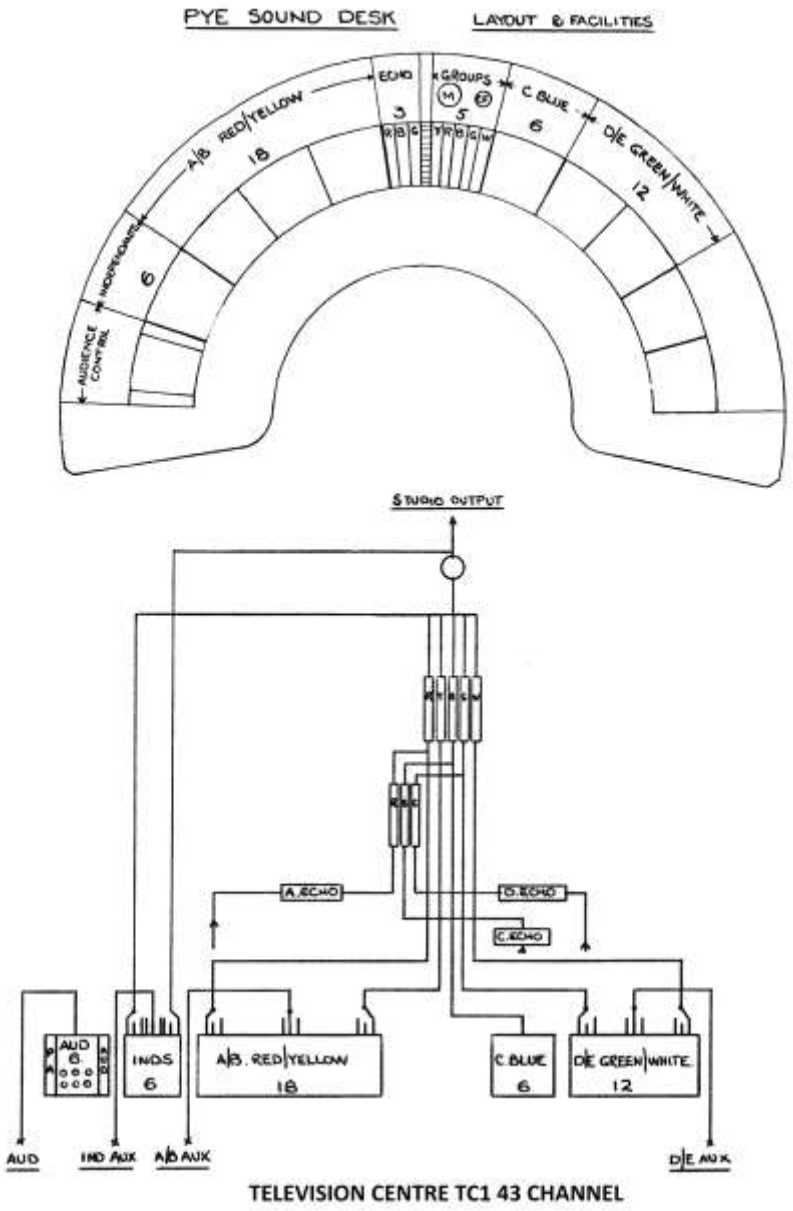


The Pye console in TC1, newly installed.
Photo: Pye audio brochure

Television Centre TC1 Sound Control, showing the view into Production, with the window to the studio floor being on the left side, behind the camera. Note that despite what I've said about the new modules in the desk, it still required the rack of output amps and power supplies visible on the right, along with yet more jackfields. The 43 channels required 11 wedges or bays which were arranged as per this drawing.

The BBC Pye Operations Manual

The BBC Head-of-Sound John Eden-Eadon wrote an 'Operators Manual' for the BBC Sound supervisors. He sketched the basic schematic of that TC1 console:



Part of the BBC Operators Manual for the Pye desk.
Via John Howell. [10]

It could possibly have been that that the usual Pye arrangement was eight Painton Faders in each channel wedge section, but like the ATV desks, the BBC's only six faders per section. The channels were assigned to the five desk groups as above, with a further six 'audience' mic amps available as a sub-mix. There were also six 'Independent' channels going straight to the output. The channels had three 'Echo' groups with corresponding returns. Echo at TV Centre was provided by a dedicated EMT140 plate in an adjacent equipment bay for each studio. You'd have to patch to another 'spare' one if you required more.

The next two TV Centre studios that got these Pye transistor 'wrap-around' desks were TC6, which like TC1 had been considerably delayed and opened in 1967. Then finally TC7 opened on 4th May 1968. [11] That's a long way then from 1961 or '62 when the first desk was probably ordered from Pye.

More from the Operator's Manual:

PYE TRANSISTORISED SOUND DESK

Installed:	TC1	TC6	TC7
GROUPS: A/B	18	12	12
C	6	6	-
C/D	-	-	12
D/E	12	12	-
Independent:	7	6	6
Total No of Channels.	43	36	30

AUD MIXER: 6 (rotary)

ECHO CHAINS: 3 (A,C,D TC7 : two only)

CLEAN FEED: 1 (any or all Inds)

MULTI-CLEAN FEED: Any of all Inds

Each Channel Has:-

- low/high switch (20dB)
- amp gain control (30dB)
- 3 pos n attenuator (±20dB) (TC1 and TC7 only)
- echo mixture (inc Inds)
- PA (backstop) switch/attenuator
- FB switch/attenuator
- Prehear (LS)
- Grouping selector (A, B or Aux)

Each group channel has Prehear (LS)

EFFECTS SWITCHING (auto/manual)

- 4 inputs to 2 EU's
- 2 aux ccts

COMPRESSION AMPLIFIERS: TC1 : 4, TC6 : 4, TC7 : 23 (2 MM.

VAR. FREQ. AMPLIFIERS: TC1 : 18, TC6 : 16, TC7 : 10

Comprehensive jackfields allow great flexibility.

Low level access: at LH side (-70dBO

High level access: at RH side (0 level)

2nd P.P.M. on desk

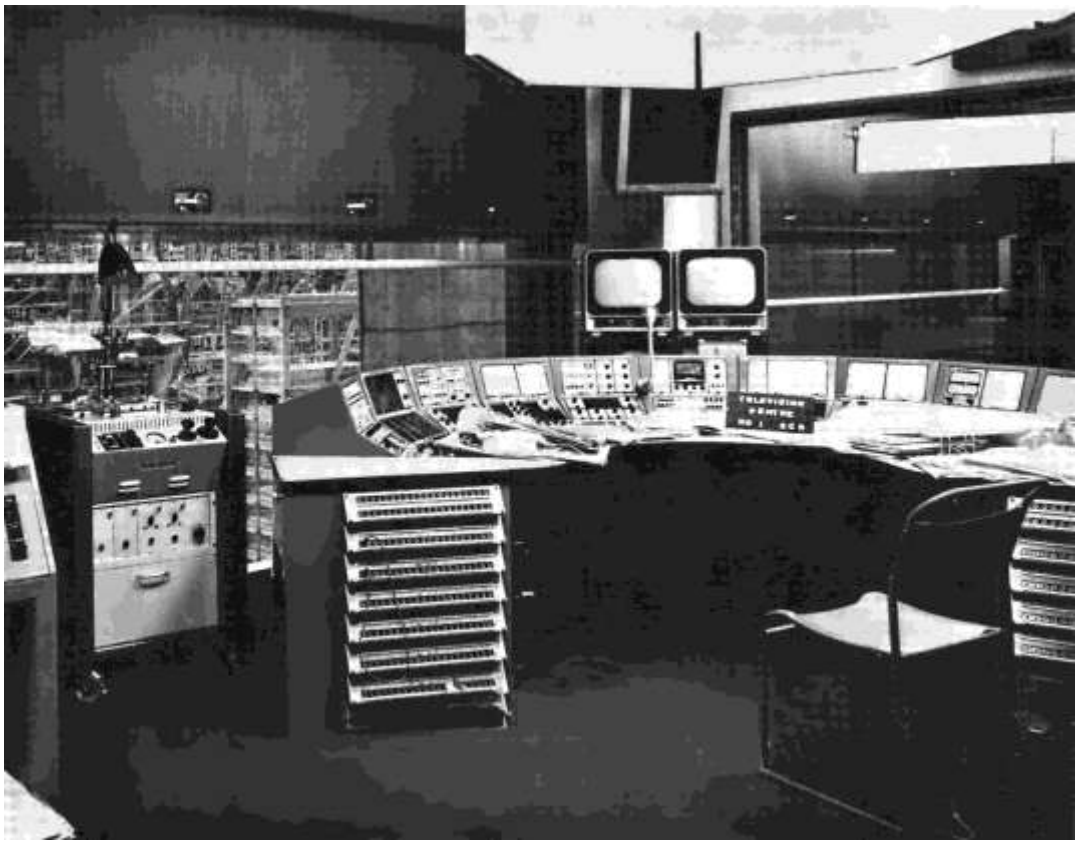
A summary of the facilities on all three of the TV Centre desks.

From the BBC Ops Manual by John Eden-Eadon – revised in March 1971 by D.Gough

The BBC Operator's Manual document details that TC1 had 43 Channels, TC7 had 30 and TC6 had 36. It states that the TC1 desk has seven of those Independent channels, the others having six and the grouping arrangements, along with the number of EQ amps and Comp/Lims, are different between them.[10]

Tweaking the monitor speaker in TC1

Here's another picture of the TC1 desk, taken in 1969, showing its location in the sound control room:



The TC1 desk in 1969.
Photo: from BBC Monograph no 78

The photo above of TC1 is from an investigation by the R&D Dept in 1969. The report was produced in response into criticisms of the monitoring in the BBC TV Centre sound control rooms, which stated it was 'coloured and tunnelly.' It does seem a long time since TC1's installation in 1964 though, during which the studio produced some fantastic sounding programmes.
The report focussed on adjusting the height of the hung LS5/2 speaker and although the best sound was having it floor mounted, that was not consider applicable here, so it was lowered until it was just above the picture monitors and pivoted forward a bit. *"The degradation in quality observed when a high-quality monitoring loudspeaker is used in a reflecting corner is satisfactorily explained by interference between the direct sound and that reflected by the surfaces forming the corner."*[12]
This also photo shows how, when TVC was built it was still considered important to have a view into the studio….even if most of what you see here are lighting hoists! The window to the right looks into the Production gallery, as shown in the earlier picture.

Note that the TC1 desk as it was in 1969 had a few missing modules…..old age coming on? A number of photos of these Pye desks show similar holes in the 'wedges', like this photo of the TC7 desk:

The TC7 desk



Nick Ware plugging up the Pye in studio TC7.
Photo: Nick Ware

Seen here in the early ’70’s, this is the 30-channel Pye in TC7. It has some of those missing modules, but it’s obvious that you could fit a mixture of the ‘Mic Amp’ modules or ‘Mic amp/EQ’ modules into those sloping bay sections.

That BBC Operators Manual has details of some of the modules:

More from The Manual

AMPLIFIER PYE 4073

Its function is to provide amplification from microphone to channel fader (low level plugging, high level mixing), group separation and line sending.

The amplifier is arranged in two sections, the input stage of 45db maximum gain which has variable controls and the output stage of fixed 35db gain.

Input Stage

Input impedance - 600 ohms - number of transistors 8

Two controls are located on the front panel.

(1) Toggle switch High/Low

High position maximum amplification

Low position 20db attenuator between the input transformer and the first transistor.

(2) The level control (molded track) giving variable attenuation of 25db achieved by the application of negative feedback.

Output impedance - 600 ohms.

Output Section

Input Z 10,000 ohms. Nunber of transistors 8

Push pull amplification. Output impedance 30 ohms.

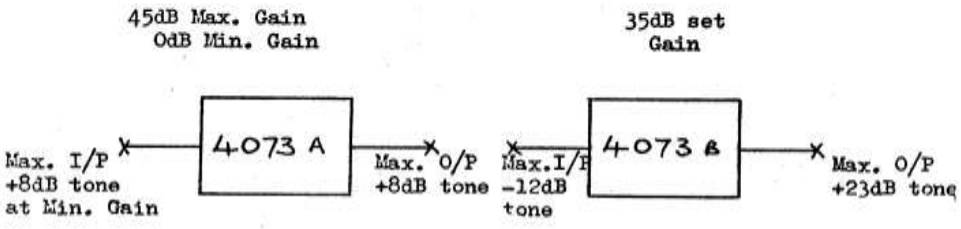
The amplifier may be used in two ways:-

(1) The two stages are connected in cascade to provide maximum gain of 80db. The operational limits are (toggle switch high 55-80db)
(toggle switch low 35-60db)

Frequency response 60c/s to 8kc/s ± .2db - 30c/s to 15kc/s ± 1db

(2) The two stages are used separately and for different purposes such as Amp No. 2. A section in Group A/B. F/B. - B section in group A C/F.

Inputs and outputs are in some cases available on jacks, the following parameters being applicable.



Details of the Pye 4073 Amp.

AMPLIFIER PYE 4073: This is a ‘straight’ Amp with no EQ, that can be used as a Mic Amp before the channel fader, or split into two sections giving a Line Amp. These are the Mic Amps fitted as pairs in some of the ‘channel’ bays. The jackfields must have been complicated if you could access both parts of the 4073’s.

PYE Amplifier 4054

This amplifier provides an overall gain of 80db and similar characteristics as the 4073. It is used exclusively as a microphone amplifier and incorporates frequency correction facilities. They are normally available as the third and fourth channel amplifiers within each wedge of six. They may, of course, be cross plugged to provide frequency correction in other facilities such as the Echo circuit. Care should be taken to ensure that such plugging is done at points in the programme chain which are similar to the inputs and outputs of the alternative unit.

The top left control provides for a 600 ohm input with attenuation of 20, 40 and 60 db. The 60 ohm input position is for low impedance microphones which is not a facility we would normally use as each microphone point has a transformer for matching a low impedance microphone to the channel amplifier.

The top right control provides for a smooth variation of 25db attenuation.

The amount of applied correction is as indicated on the sloped on page 3, the gain and attenuation being of two standards, that of 6db and 12db per octave.

PYE Amplifier 4065

Its function is as a distribution amplifier or as a "buffer" amplifier with low gain.

Input Imedence 10,000 ohms.

Gain 20db. There is a screwdriver pre-set gain control on the front panel which operates between the input transformer and the following amplification stages.

Frequency response 60 c/s to 8 Kc/s ± 0.2db
30 c/s to 15 Kc/s ± 1db

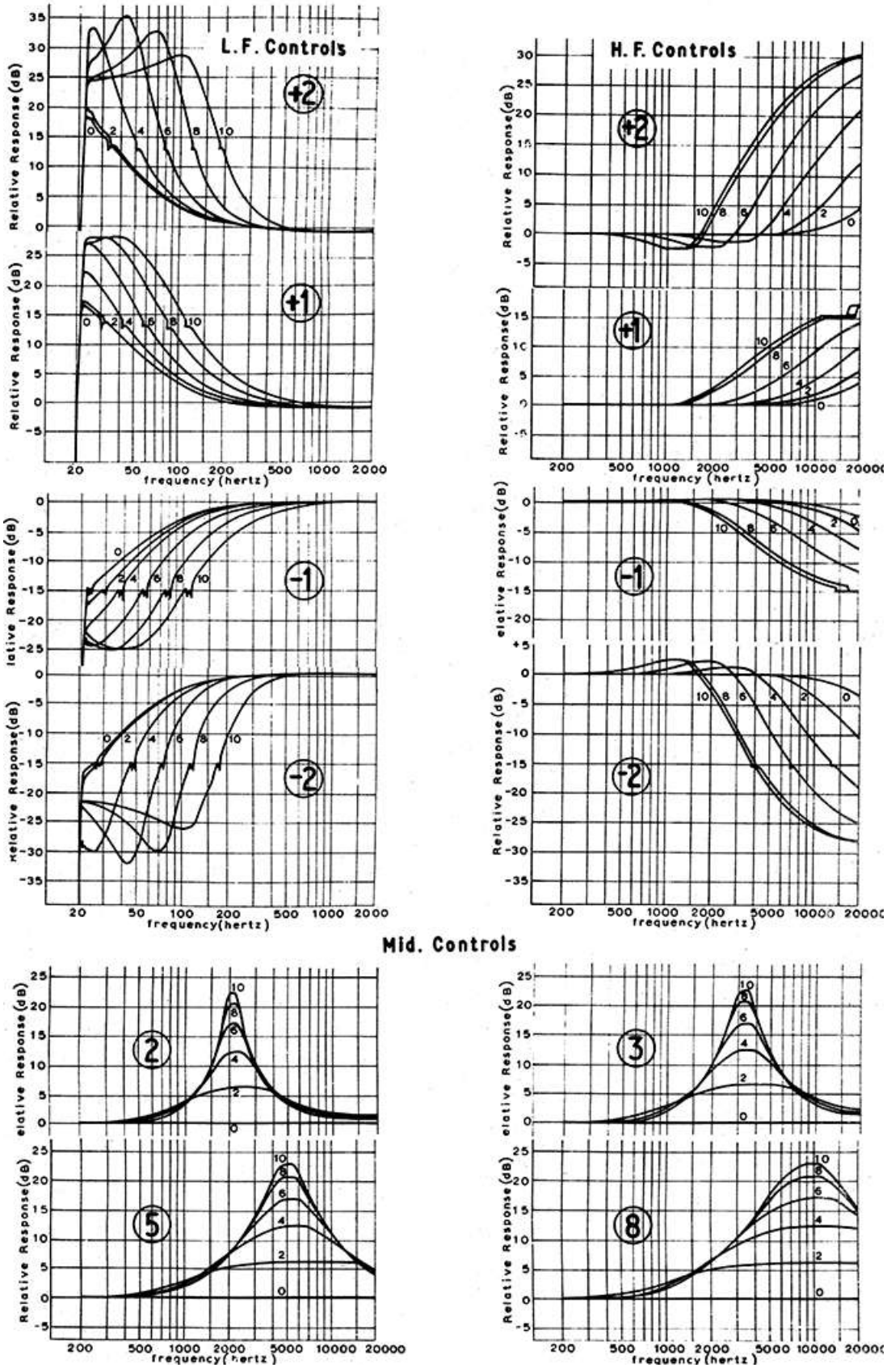
Output Impedance 50 ohms

Details of the 4054 and 4065 amps

AMPLIFIER PYE 4054: This is the 'Mic Amp with EQ' module. It states that these amps are usually fitted "as the third and fourth amplifiers in each wedge of six". As the 4054 is twice the depth, it follows that with 2 in a wedge, then 4 more of the 4073's would complete your 6 fader requirement. You just had to know where you wanted that 'Eq'.
AMPLIFIER PYE 4065: Just a 'buffer stage' or SDA, so that'll be what we see so much of in the rack beside Nick Ware's left shoulder.

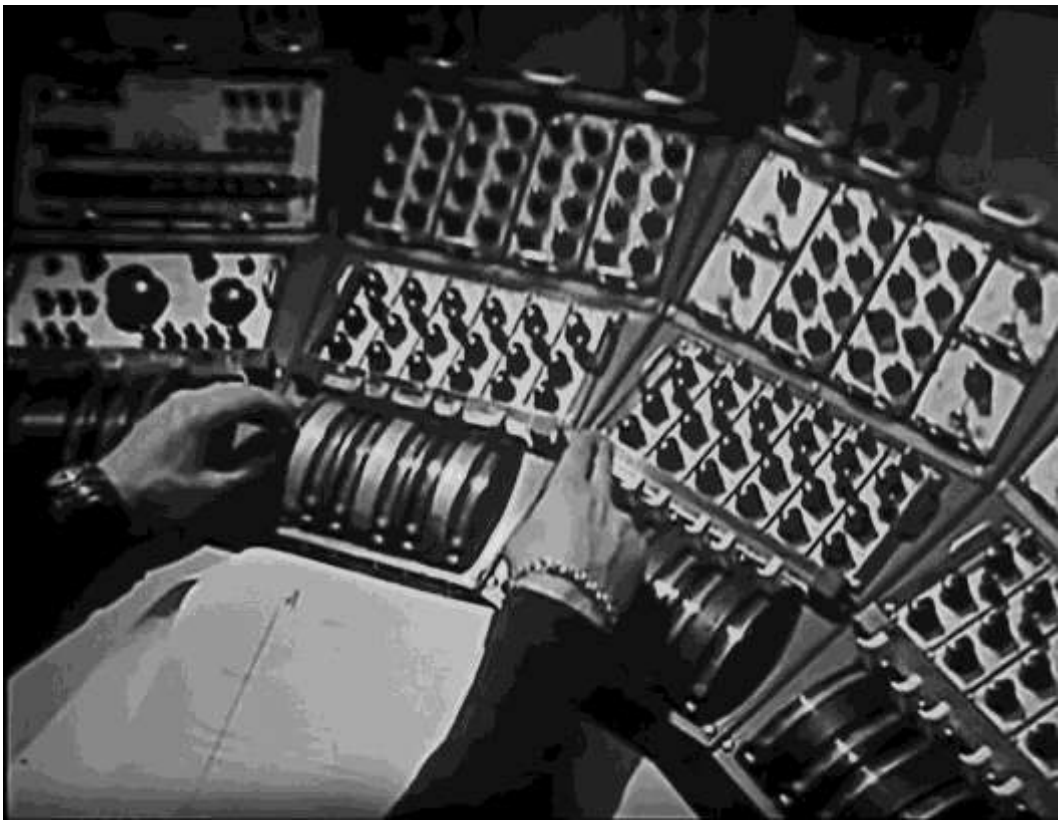
Let's see just what that 4054 EQ unit, or 'Response Correction Amplifier' as the BBC were perhaps still calling it, can do:

PYE Amplifier 4054



The 4054 module EQ curves

The Pye Mic Amp/EQ module 4054 had two sets of Low Frequency curves, either a 6dB or a 12dB per octave slope of lift or cut at a fixed frequency, which looks like 50Hz and a matching set of High Frequency curves, 6dB or 12dB up or down based at 10K. The Mid Frequencies are 2k, 3K, 5K and 8K in steps of 2, 4, 6, 8 or 10dB up or down.
Finally then the Beeb Sound Supervisors had EQ on many….but not all their desk channels, without resorting to external units. The manual shows that in TC1 you could have 18 of your 43 channels with EQ, in TC6 then it was 16 of your 36 and in TC7 on just 10 of your 30 channels.



Here's a lo-fi 'screen grab' from a Blue Peter of 1974 showing the TC7 Pye desk.

This almost vertical view shows the operator's left hand on the 'Echo and Group' faders. Being the TC7 desk, we know from the spec sheet that it had two 'Echo Groups' and four 'Programme Groups'; therefore six faders. His right hand is on that bank of six faders and the six 'Auxiliary' cassettes are just above them. Next up, comes the section here populated with four of the 4073 'Mic Amps' and two of the 4054 'Mic/EQ Amps'.
Finally the top section, hidden in the darkness are the panels with additional facilities. Lots of space for the script there as well.

THE PYE 4060 COMPRESSOR/LIMITER

As I said in my earlier articles on the Pye OB desks, the Pye 4060 Comp/Lim has remained the most lasting of the products from the Pye Audio Dept under Bob Barrass.

It is still a highly sort after piece of equipment and when I last looked there were a 'pair' for sale online at just under £14,000….60 years after it was designed.

Here's a modern pic of a 'single unit':



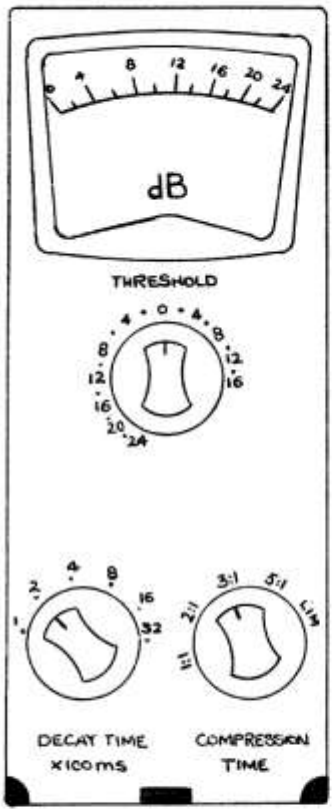
A nice clean looking 'mono' 4060

So what's all the fuss about this 60 year old, transistor design; it would have been 'germanium' transistor when developed, although perhaps 'silicon' in later versions.

John Eden - Eadon, who's surname was pronounced 'Eden - Eden' and I hear was therefore knicknamed 'John - John', once again tells the story best in his BBC Ops Manual:

GENERAL SPECIFICATION

DIAGRAM I



Control Range

This amplifier provides for 5 operating conditions.

1. A linear position for setting up associated equipment.
2. Compression Ratio of 2:1
3. " " " of 3:1
4. " " " of 5:1
5. Limit condition in which the ratio of input to output is greater than 20:1.

Frequency Response

Under all operating conditions response is maintained within 1½dB variation 30Hz - 15KHz.

Attack Time

Attack time is fixed under all conditions. Compression-less than 0.5 m.secs. Limit-1 m.sec. ±0.5 m.sec.

Decay Time (Recovery)

Decay time is switchable to 100, 200, 400, 800 m.secs. and 1.6, 3.2 secs.

Threshold Level

Threshold control operates in 2dB steps over a 40dB range, 24dB one side and 16dB the other side of a normal operating position.

Nowadays that quoted frequency response spec of within 1.5db may rise a few eyebrows if paying 'modern money' for it, I guess. You'd have to 'squeek it' to see what the frequency response was really like I think. Note also that this model, as used by the BBC in their studio Pye desks, does not have the 'Noisegate Threshold' control shown in the modern photo above. There were many versions of 4060 I'm sure, as typified by the images of them in various single and dual rack models. The ATV ones that were being delivered from 1961 had 'four knobs', so I possibly had the Noisegate back then, but perhaps the Beeb just didn't feel the need for a 'Noisegate'…not a very 'broadcast thing' after all.

That Mike McCarthy sound

Mike McCarthy was one of the BBC Sound Supervisors newly made up in the days when the TV Centre studios had either the Pye desks or the BBC Type B's. Mike became renowned for his very exciting sounding 'audience'…..many of us mixing wanted to emulate his 'up-front' audience and well controlled dialogue on 'Sit-Coms'.

Mike remembered:
"You could tell which were done in TC1 by the closing applause, because I loved the Pye Limiters, they really 'scrunched it up' so beautifully. Whenever I see the odd repeat I think "Oh yes that was TC1 because the end applause was so 'squeench' (Mike makes the noise of distortion!), whereas the other studios had different limiters. Certainly in TC1 and with the sort of acoustic in the studio, they really did stop anything. They were beautiful but they did give you this very harsh sound, which was very popular at the time." **[13]**

The Manual also stated:

PYE COMPRESSOR/LIMITER AMPLIFIER 4060

This unit was designed to BBC specifications initially for use in the Pye sound desks in T.C.1 and Television Theatre. Later models with code number Pye 842678/00, BBC No. LF104 were modified to suit the BBC DK4 and EMI Sound desks. These are incorporated into standard cases which include RSA's of the BBC type. The latest sound installation in T.C.8, the BBC Type D desk, employs the original 4060 and only requires a different threshold setting to accommodate a lower input level. The normal threshold is approximately 8 dB higher: - 8 on the threshold control.

So from that note in the Manual, which got amended in 1971, it looks like these are as specified for the first two of the Pye desks, in TC1 and the TV Theatre. The DK4 mentioned was a fairly basic design, often built as 'radio presentation' type desk and the Type 'D' was the last of the BBC designed desks which I think used commercial EMI Broadcast modules and one went into TC8 and another also replaced the Pye in TV Theatre in the early '70's. It carried on the 'wrap-around' layout and small Painton faders that had been introduced by the Pye consoles.

So for the guys out there now prepared to cough up £14,000 for a pair of 4060's...let's look further into that interesting Compressor/Limiter design, and the technical bits of how it worked:

General Description

The unit consists basically of three sections:

1. The input amplifier.
2. The compression/Limit attenuator.
3. The output amplifier.

The input and output amplifiers together, always provide the same gain, the threshold control operating as a pair of opposed attenuators, one preceding the input amplifier and the other controlling the amount of feedback over the output amplifier. The overall gain is always unity and therefore allows for ready insertion into different parts of an installation without requiring compensating adjustments in the particular programme chain.

The unit functions in the normal way of a compressor/limiter. It has a main programme chain, a control element and a side-chain developing the control signal.

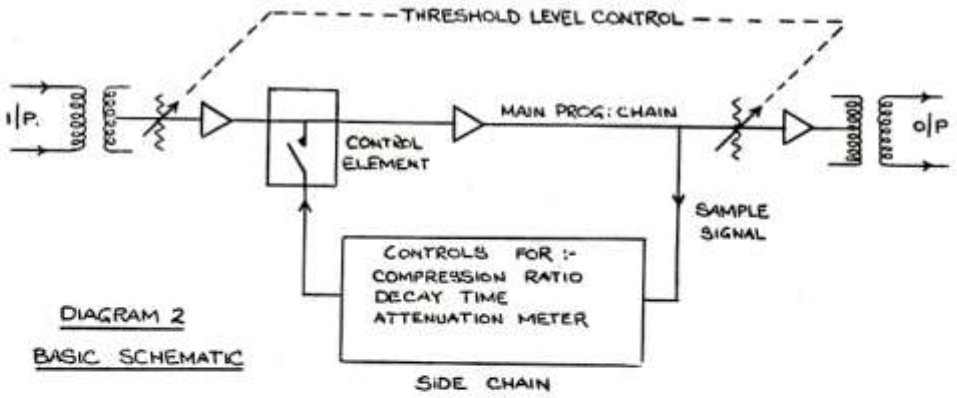
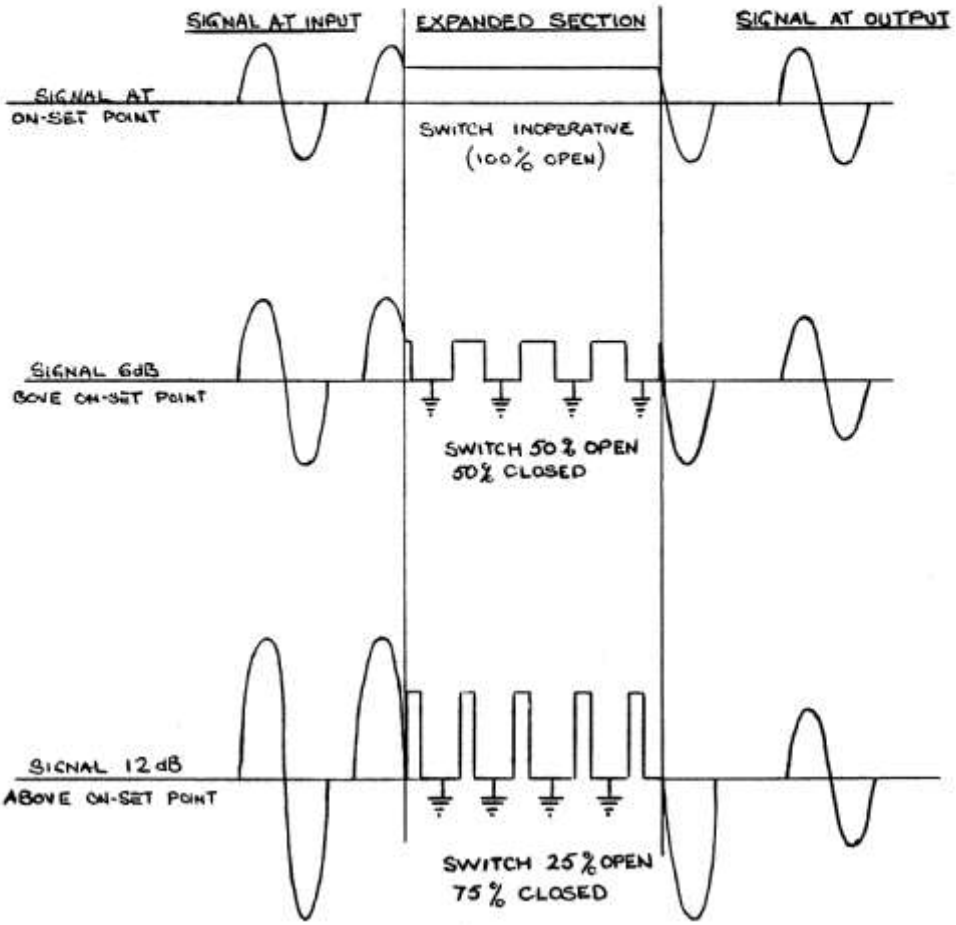


DIAGRAM 2

BASIC SCHEMATIC

The side-chain samples the programme and processes it to produce a variable amount of D.C. control signal depending on the mode of operation. The control element in the programme chain is a transistor switch, chopping the audio signal at 250 KHz with a variable mark-space ratio pulse. When the signal is below threshold, the mark-space ratio is increased to the limit when no chopping occurs.

For increases of signal above threshold, the signal line is earthed for progressively longer fractions of the periodic time of 4 microseconds. The effect of operating the switch in this varied manner is to remove some of the excessive energy contained in the original signal, the greater the width of the gaps the greater the amount of energy removed, and therefore a reduction in the dynamic range is achieved.



So, thanks to John Howell's diligence in keeping his BBC Operational Manual notes for so many years, we now know how the BBC Pye desks were configured and how the 4060's worked.

Here's my last picture of a Pye desk at Television Centre in July 1969:



I'm not sure what unit the engineer here is touching, not a 4060 but it's got a meter on the face though.

Apollo 11 Splashed down on 24th July 1969. The BBC were not usually involved in advertising, but this photo was used in an ad for Ferrograph 7 tape decks in the January 1970 Studio Sound magazine. The Ferrograph is positioned in an 'unlikely' place beside the Pye console, as all tape decks would surely be in the 'grams area'. The studio beyond has a pair of EMI2001 cameras visible, so it's one of the new colour studios with the EMI's, either TC1 or TC7.
The photo once again illustrates how the 4073 mic/line modules were liberally spread about in the upstand wedges in the Pye desks.

The Television Theatre desk

Now let's look at another of the big Pye desks put in service by the Beeb, the desk at The Shepherd's Bush Television Theatre. The delay with Television Centre's studio TC1, probably put this TV Theatre desk into service first, so can anyone confirm the date of its introduction for me, as I believe it was 1963?



Adrian Bishop-Laggett in the TV Theatre.
Photo: via Bob Foley and the Tech-Ops website

Adrian Bishop-Laggett, nicknamed 'Bish', another of the great BBC Sound Supervisor's, at the 36 channel Television Theatre desk, probably in the early '70's.

From The TV Theatre in 1963.
"An Evening With Nat King Cole" ....in colour

Let's take a look at an one of the many 'one off' shows made at the TV Theatre, and one surely mixed on the Pye and done in the Theatre in July 1963 – *"An Evening With Nat King Cole"*.
The sound is absolutely superb and without any 'credit for sound', the bets are on this being mixed by Hugh Barker. However you'll notice that it's in colour….but that's surely impossible as the cameras in TVT in 1963 were monochrome Marconi Mk3's.
The answer lies in the very final end credit on this American 2002 DVD which says *"Digital Color by CST Entertainment Inc"*….yes it was 'digitally colorised' from an original videotape.

It does look superb, so 'well done' CST Entertainment, but moving on from the cleverness and quality of the colorising….what about that sound. As I said, it surely was a Hugh Barker show and it undoubtedly has a fair amount of 'miming' to Hugh's pre-recorded tracks. But note at the beginning of this clip, the first transition from speech into 'Unforgettable', where the music comes under Nat's speech and the tightness of the shots throughout says that's a

‘live vocal on a boom’ to me. The next two songs ‘Day In, Day Out’ and ‘Here’s That Rainy Day’, I’ll will put down as mimed as I don’t really see a boom coping with the shots and lighting. For awhile though I decided it was all live and that he was on an early BBC radio mic, however I’ve been told ‘not at TV Theatre in 1963’…so I’ll say that Nat’s lip sync is remarkable.
You had to keep the foldback speaker surprisingly close to avoid any added ‘playback delay’ for the artist lip-syncing, so a member of the crew usually tracked the big wheeled speaker cabinet in and out, avoiding the cameras and their cables of course. The artist had to precisely remember their recorded singing of course!

TO VIEW THIS VIDEO PLEASE SEE THE ORIGINAL ARTICLE AT: www.postfade.co.uk

I’m impressed with the 2002 digital techniques produced in making this amazingly good colourised version of the 1963 BBC programme.
Excerpt from Image Entertainments DVD

The boom op would here be working hard avoiding that very ‘frontal’ follow-spot on the speech and the live vocals and the gram op also had some frighteningly tight ‘cues’ to play in. All in all a wonderful example of the Beeb in 1963. The Director was Yvonne Littlewood and the good stage design by Stanley Dorfman.
The band, by 1963 was now tucked away in a ‘band room’ at the TV Theatre, I believe, and is credited as ‘The Augmented Ted Heath Orchestra’. There’s a decent number of string players in there, all being conducted, not by Ted but by American Joe Zito. With the ‘simple but effective’ production and the great sound, I can see why an American production company would pay for the ‘colorising’ for a DVD release in 2002.

‘The Black and White Minstrel Show’ from the Theatre…and Lansdowne

One of the long running BBC shows made at the TV Theatre was ‘The Black and White Minstrel Show’, a weekly peak-time programme which ran for 20 years from 1958 with massive audiences.
In 1961 it swept the board against international competition at the European Broadcasting Union’s television light entertainment festival in Montreux, winning the Golden Rose, the Silver Rose, and the Press award for the best show. It wasn’t until 1967 that the BBC were petitioned to close the programme for its grotesque racial stereotyping, but despite attempts at alternate, less offensive versions, it returned and ran until 1978.
It seems surprising that 14 million Britons frequently watched this show without understanding the hurt it gave to our black citizens. It is now widely regarded as an embarrassment. [14]

Below is the Producer George Inns with his Production team in the Television Theatre Production Gallery, with Sound beyond.



The Television Theatre Production Gallery with left to right: TM Dave Sydenham., Vision Mixer Gladys Davies, Producer George Inns and PA Pat Molloy.
Seated at the desk in Sound beyond is Sound Supervisor Adrian Stocks with standing Eric Wallis and Trevor Webster.
The photo was published in the 1962 BBC book 'The Black and White Minstrel Show'.

That photo taken in 1962 possibly pre-dates the arrival of the Pye desk in the Theatre though.
The principal singers on the Black and white Minstrels were often live, but singing over pre-recorded tracks…..tracks that weren't recorded by the BBC, but at Lansdowne Studios by Adrian Kerridge. The shows musical director George Mitchell believed he needed the same engineer throughout the shows, something the BBC couldn't guarantee.
Kerridge tells the story in his autobiography: *" I was requested as engineer and recording the show entailed recording all the music, twelve male singers, of whom three were also soloists, including Dai Francis, who could sing in the style of Al Jolson, plus John Boulter (tenor) and Tony Mercer (bass baritone). the Minstrels also included eight girl singers – "The Television Toppers" – with Margaret Savage solo. There were guest artists including George Chisholm, Stan Stennett, Leslie Crowther and tap-dancer Benny Garcia, tap-dancing away on (one instrumental piece) a tap mat in the small separation room next to the control room. A regular guest with an impressive voice was Gerry Dorsey. In the mid 1960's he assumed a stage name of Engelbert Humperdinck."*
"The band consisted of 20 (it sometimes varied) musicians all recorded in a

four-hour morning session: not an easy task with only a 12-channel input board. I used one Neumann U-47 on the men, who were raised on rostrums. On the girls, who were formed in a circle, a Neumann SM2 using both the mic capsules in a wide cardiod with the top capsule rotated 180deg from the bottom one. A Neumann U47 was used for the solo singers. Tall screens were used to separate the singers from the band and the songs were recorded in separate segments.
After the lunch break, I had to edit the show making any musical performance edits required and some titles into extended sequeing. This usually took two to three hours, depending on the complexity of the show. The show was recorded directly to mono, with no multitrack. A copy with no edits was made. The BBC didn't want any edit to fail on live transmission."
An interesting note from Adrian also was that he copied the tapes 'backwards', as that retained the transients. [15]

Pye desks… on the move

When the TV Theatre was being converted for colour, with new control rooms, another theatre, the Hippodrome at Golders Green was pressed into use. During this time in 1968/9 it was equipped with a Pye, which would have been the TV Theatre one temporarily moved:



Pye console in the Golders Green Hippodrome
Photo: Barry Bonner

Barry Bonner took his new Praktica 35mm camera along to the Golders Green Hippodrome on Saturday 29th March during a Rolf Harris Show, to take these great shots. Rolf Harris was at Golders Green from January to the end of March. On the left side desk is an old MX18 passive mixer, obviously for more

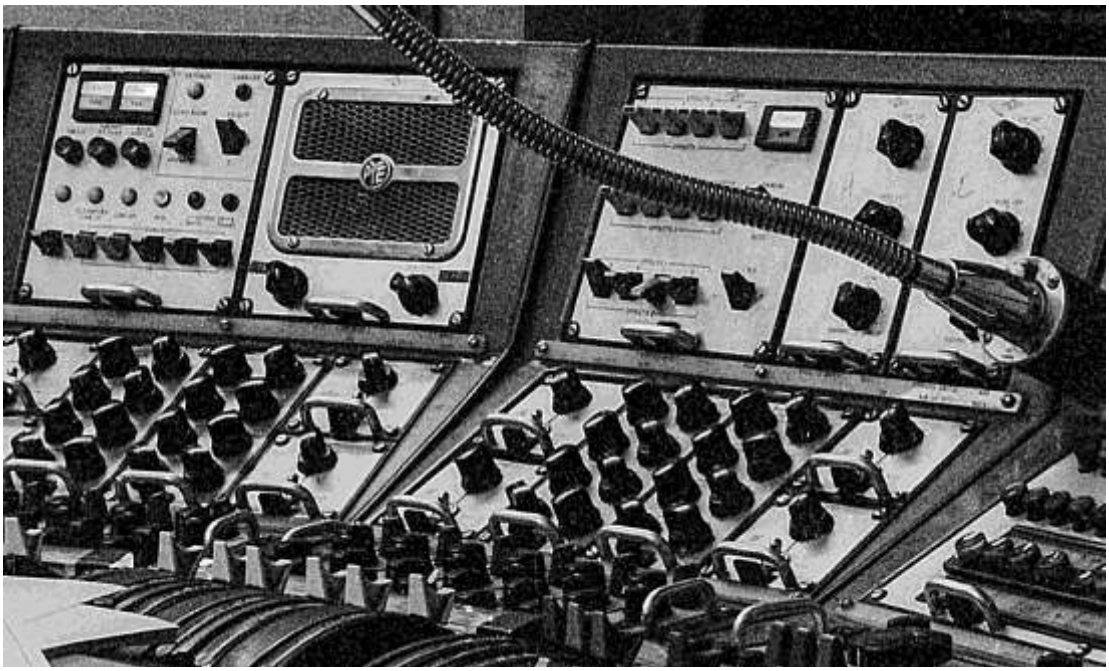
audience mics, oh…..and a can of switch cleaner.
Barry's second picture enables us to get a closer look at the centre section:



Taken during the recording of a Rolf Harris Show on Saturday 29th. March 1969 at The Golders Green Hippodrome.
Photo: Barry Bonner

It was an extremely tight fit in the small sound control room and even the EMT140 Reverb Plate had to be in there as well. It was normal to keep the EMT's away from any risk of vibration, but that's part of the reverb unit's 'wooden crate' housing which can just be seen in the top left of this image. The faders in the centre here, are the 'Group Faders' and there are 'Main' and 'Clean Feed' rotary pots in the section above them. Then comes a set of selectors on a panel. To the left of that, with the talkback mic passing in front of it, is the 'Telephone Effects Panel'.

Telephone Distort Effects



On the upper right is the 'Tele-distort Effects' panel, with on the right, the two EQ filters that also produced the degree of 'distort'.

Close up we see the pre-selector switches to facilitate 'telephone distort' switching on dramas. During a TV Studio Drama it was pretty common for some important plot dialogue to take-place by way of a telephone call. The cameras chosen by the director to shoot the scene would be identifiable from the camera script, and the Sound Supervisor would use this pre-selector panel to assign those cameras, when 'cut' by the Vision Mixer, to then switch the distort effect in or out for the 'other side' of the phone conversation.
The actual EQ units are on the right of this Telephone Effects Panel, and varying degrees of EQ and distortion can be chosen, depending on the quality required for the voice on the phone. The sound coming from a couple of studio booms over the actors concerned was of course the sound being 'distorted' though the Effects Unit.
Confused? Well here's an example of a 'tele-distort' sequence from a drama. I'm really sorry I couldn't find a BBC drama at all, or one mixed on a Pye, but it illustrates the device:

An excerpt to explain 'Tele-distort'

TO VIEW THIS VIDEO PLEASE SEE THE ORIGINAL ARTICLE AT: www.postfade.co.uk

A 'telephone distort' sequence from a Callan episode. It was a Thames show….and would have been mixed on one of their Neve desks by then, however the 'distort' units were of similar concept.
The 'distant voice' on the telephone call is heard distorted when the camera has cut away from the person speaking. A cut could happen during the phone dialogue as well, but this was usually avoided.

In the above example, the phone is perhaps a bit too loud when the street names are being given….she would surely be hearing that as well! Just how low to drop the 'distorted voice' was always a difficult judgement.
I also recall that since both actors making the phone call were in the studio at the same time, you had two booms faded up during the sequence. If the sets were close, or either of the actors got too loud, they could acoustically 'spill across' onto the other boom, which completely destroyed the 'distort effect' that you were nicely keeping low in the mix.

A TV Centre Pye goes to Birmingham

In 1971 the BBC Birmingham TV and Radio centre at Pebble Mill opened and the TV Studio got one of the London TV Centre Pye desks after refurbishment. A BBC report stated:
"The sound control facilities are provided by a Pye desk transferred from London. This transistorised equipment, containing thirty-six channels, has been modified to give group switching facilities. Each channel can be routed to any of three destinations – either of two groups or auxiliary – and each group of six channels can be switched to main clean feed, main output or independent. Auxiliary facilities include sixteen R.S.A.s (Response Selection Amplifiers) and three echo chains and a separate six-channel mixer for audience use." **[16]**
The Report says "36-channels"….so that was the ex-TC6 desk then. I wonder how long it continued in service in Birmingham?

THE PAINTON QUADRANT FADER

The valve desks had the 'big' wide Painton Quadrant faders, that had two rows of 'studs', that the 'wiper' passed over to achieve the attenuation and the new transistor desks adopted Painton's newer smaller quadrant introduced in 1959, as here:



Developed in 1959, the later smaller Painton Quadrant faders, with the wonderful illuminated bezels.

‘Quadrant’ because in the design the wiper was connected to a pivoting arm that traversed a section of a circle…’the quadrant’, as it was operated. It was the easiest way to build a ‘bridge ‘T’, stepped ladder’ attenuator that could remain at a constant impedance whilst varying the signal level. Impedances were matched throughout in audio circuit designs, as for many years the broadcast industry followed the telecoms world in using the same impedance for the ‘input’ to a source as was used in the ‘output’ of the previous section. This gave the best transfer of the signal without loss and it had been settled on as being 600 ohms.
This had however worked well in the Post Office, with very, very long lines but when the recording world followed this pattern, it only made sense if you had say ‘input’ such as one tape recorder with an input impedance of 600 ohms, connected to your 600 ohm ‘output’ amplifier. If you now connected a second tape recorder, also with a 600 ohm input, you now got a 6dB loss in signal. Once you started getting more ‘inputs’ to be fed, a better way was to use a low output impedance and then have high input impedances on the following equipment. The overall losses were less. Once that was established even the faders got changed to being 10k ohms instead of 600 ohm.
The 600 ohm Painton faders had 31 steps. From 0 to -15dB these were in 1dB increments. The taper beyond that was as follows: 6 steps at 2dB per. Followed by 3 at 3dB, 2 at 4dB, 1 at 5dB, 1 at 7dB, 1 at 9dB and off. [17] These were in ‘steps’ and not continuous, which didn’t matter on almost every audio signal, but even these small steps could be heard on constant tones….or even a held note from an organ perhaps.
As you can see in the above photo, the later Painton quadrant came with a wonderful illuminated bezel and an internal micro-switch which switched on an additional light when you lifted the fader off the backstop; I don’t think faders have ever been as beautifully ‘visual’ ever again.

I do find it rather bizarre though that a fader designed in a curved shape, as a way convenient way to build a constant impedance 600 ohm volume control, could still have a life so long after flat audio faders were introduced and became the standard everywhere.
BBC faders of course were ‘pulled’ towards you to fade up and it must been the way your hand felt when the faders were ‘up’ and toward you, that made them feel just ‘right’, because BBC TV Sound Mixers were so enamoured of the quadrant fader, they even insisted they were used in two SSL 5000’s mixers installed into their new TV Centre dubbing suites Sypher 3 and 4 as late as 1987, and they also had them motorised! These installations were therefore the first to use of motorised faders in SSL consoles….and caused endless difficulties to accomplish I believe.[18]

BBC Radio’s Pye desk

There was one more Pye that I know of, in the BBC Radio Studio at the Camden Theatre, used for music programmes, which was also the home for the BBC Concert Orchestra until 1972:



Mixing 'Friday Night Is Music Night' at the Camden in 1965.
Photo: BBC

This Camden Theatre installation was the only Pye console in BBC Radio, installed in the 1960s. Note all those empty Mic Amp sections on the right hand side in this picture.
A fascinating history website written by ex-BBC Radio guys has a long piece on this desk, which I do hope they don't mind me quoting from: [19]
"The Pye console was built in a semi-circular format and had four 8-channel groups and four independent channels. Each set of 8 channels was separated from the next by a wedge-shaped spacer; a centre section housed the group and independent faders. The master fader was a large rotary knob, with a second identical one alongside designated as the standby output. The faders were quadrant type, and worked in the opposite sense to those in the commercial studios of the day, fading up towards the operator. This enabled the fader graduations towards the open end to be seen, as the curvature of the faders meant that the operator could not see the other end. Line up, with 10dB in hand, was stop 23; the graduations went from 0 to 30 but in a linear manner. There were two associated rotary controls for each fader, echo send and PA send. The echo control was a dual log/antilog potentiometer providing echo mixture in a similar way to the standard BBC echo mixture switch but continuously variable. This was later used as a pan pot when the desk was adapted for stereo, which was simply done merely by using the echo send. Each channel had its own RSA – Response Selection Amplifier – the BBC term for what was known elsewhere (inaccurately) as an equaliser, and there were several (four?) compressor/limiters which could be patched into channels or groups. There were two enormous jackfields, one at each end of the console built into the ends of the semicircle."

"The electronics were based on germanium transistors, which were highly sensitive to overheating. The console was cooled by a forced air system with

the fan unit in the corridor outside the cubicle. A large orange lamp indicated that all was well, and it was rumoured that if the fan failed and the lamp went out we had about 30 seconds to power the desk down before it melted in a mini China Syndrome! The transistorised electronics was also quite noisy, with a very small margin between overload and excessive noise, and there were strict instructions about the permitted usable range of the main fader, which was about 10dB!"

"For audio monitoring, two loudspeakers were provided – an LSU10 for the Popular Music SMs [Studio Managers ie mixers] and an LS5-1 for the Light Music SMs. Due to the large size of the cubicle it was difficult to achieve a satisfactory listening level with either of these units so they were often used together by the Pop Music SMs. To ensure that any distortion was not caused by loudspeaker overload the practice was to send tone at PPM6, set the desk LS control to maximum and turn up each LS gain control in turn until distortion occurred, then back it off a little. That was quite loud!"



***Chris Chambers preserved the 'Camden desk' when it was replaced.*[23]**

The Camden Pye desk was saved by Chris Chambers and in his picture above, there are 'stereo bars' strapped to some faders, thus joining fader pairs, and the key switches between channel faders that were on the BBC desks can be seen. Certainly 'Friday Night Is Music Night' would use stereo C-24's mics for the main orchestra pickup.
How fortuitous that because the 'Echo' pot was in fact a dual log/antilog pot, it could easily serve as a 'stereo pan-pot'. I assume you took an Echo Output and made that 'Left Output', and the desk Main Output as 'Right Output'…and you had a stereo audio desk; brilliant!

BBC Radio Jazz Club at the 'Camden'

Another regular music programme recorded at The Camden was 'Jazz Club'. Here's a part of an article about making one of the Jazz Club programmes with

the sound mixer Joe Young, published in Studio Sound November 1973. Joe was recording 'as live' of course, the Stan Tracy Big Band and The Spontaneous Music Ensemble.

Adrian Hope writes:
"The mix that goes into the tape room comes from the control box where Joe Young rides a Pye 36-channel console for Jazz Club producer Ray Harvey. On the night I was there, only 13 of the 36 channels were being used but sometimes, especially when the Camden is used for light orchestral work, all 36 channels are opened out. Sometimes also an eight channel sub -mixer is used for the strings to provide an effective total of 43 channels. PPMs are used on the Pye console, one each for left and right, with VUs on the Studers for lining up. Monitoring is through two standard BBC monitor speakers with provision for muting to half volume (the normal monitoring level is high, well above the auditorium sound level) and there is also provision for listening in mono.
There is no limiting because as yet ganged compressors are not available and, unless the compressors are ganged, peaking on one channel only will obviously cause that channel to be sat on and the stereo image to shift. In the control room there is another Studer, also running at 38 cms which provides a constant short delay signal feed to a pair of reverberation plates. This delay-plus-reverb is left on constantly throughout recording and adds to the natural acoustics of the hall".
"Humph (Humphrey Lyttelton) thanks the audience for coming along and asks them to applaud when they want, but not to whistle. 'The people at home have spent a fortune on their stereo receivers', he explains, 'and if you whistle they will think their valves have gone'. The audience duly applaud and the Stan Tracey Big Band roars into action. Their kind of music has a magic all its own. Controlled light and shade and dynamics generate a sort of excitement that is peculiar to big bands. A feeling of power in reverse. The sound coming over the BBC monitors is the sort that runs shivers up one's spine. After a few minutes it suddenly dawns on me. The reason why Joe Young gets what he does out of a band is quite simply that, once he has set up the mics his way, organised the musicians round his set-up and done a basic balance, he just doesn't touch anything. Apart from keeping an eagle eye on the meters in case something sticks their sax bell right up over a mic and heavily overloads it, he never alters a fader setting. In fact, the only settings that go up and down are Humph's voice mic and the audience reaction mic." **[20]**

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Credits and References:

[1] I haven't found details of any earlier sound consoles using a truly modular design throughout than these 1960 Pye desks. Siemens in Germany certainly had a desk with plug-in modules containing Neumann V72 valve mic amps, but not in such a comprehensive way as the Pye's. Anybody know of anything earlier?
[2] Conversation with Dave Langridge, Ex-ATV sound crew recorded by the author 6th March 2021.
[3] Confirmed by the Transdiffusion website- 'This Is Elstree Part One'
https://www.transdiffusion.org/2010/10/04/this_is_elstree_1
[4] Some episodes of 'The Plane Makers' are available on DVD from Network.

[5] From Transdiffusion: https://www.transdiffusion.org/2016/10/13/these-are-the-plane-makers
[6] 1963 was the year that Ampex developed the 'Editec' and the start of electronic editing but I doubt if it had arrived at ATV when this was made. Anglia TV was still cutting 2″ QUAD tape when I joined in 1966.
[7] Correspondence with ex-ATV soundman Richard Knight, May 2021.
[8] Details of Television Centres opening dates from Martin Kempton's great all encompassing website '*An incomplete history of London's television studios*': http://tvstudiohistory.co.uk/
[9] From: http://downloads.bbc.co.uk/historyofthebbc/tvc_handbook.pdf
[10] These parts of the BBC Operators Manual for the Pye desk, originally written by John Eden-Eadon, and revised in March 1971 by D. Gough had been 'saved for history' by BBC Sound Supervisor John Howell.
John Eden-Eadon was Head-Of-Sound when the desks were being specified in the early '60's. A very 'old school' BBC type, and because his surname was pronounced 'Eden-Eden', Keith Warren remembers that he was referred to as 'John-John'!
[11] TV Centre dates for TC6 of 1967 and TC7of 4th May 1968
from: http://www.tvstudiohistory.co.uk/tv%20centre%20history.htm#tc1,%206,%207
[12] From the BBC Engineering Monograph No78-Part Two sept 1969: "Monitoring-loudspeaker Quality in Sound Control Rooms" -by Harwood and Gifford.
[13] Conversation with BBC sound Supervisor Michel McCarthy May 2012. He made a considerable name for himself mixing 'sit-coms' and we at LWT regarded his 'up front' audience and dialogue very highly in giving an exciting audience show sound.
[14] If you're one of those who still think the series was 'rather harmless', there's a well considered look back at the offence that was caused to black people in the UK by 'the Black and White Minstrel Show', called 'Time Shift' was shown on BBC in 2005, and a short section of video from the programme is still available on the BBC History website:
https://www.bbc.com/historyofthebbc/100-voices/people-nation-empire/make-yourself-at-home/the-black-and-white-minstrel-show
[15] Adrian Kerridge sadly only managed to complete 'Part One' of his Autobigraphy "Tape's Rolling, Take One", published in 2016 by MY Books. It is well worth reading to discover about this great engineers early days at IBC with Joe Meek and his building up Lansdowne Studios and the jazz and mood album work he became so famous for.
The 12 input console mentioned by Adrian in the quote was the EMI valve console built by EMI to Joe Meek's specification in 1959 when Lansdowne first opened.
[16] Details of Pebble Mill from the BBC Engineering Information Report No.81, dated July 1971.
[17] Details of Painton Quadrant fader from: https://groupdiy.com/index.php?topic=33141.0
[18] BBC Report of Autumn 1987 on Sypher 3 and 4: http://www.bbceng.info/Eng_Inf/eng-inf-files/EngInf30.pdf
However details of the fitting of Quadrant faders and photo were posted by John Howell on the Tech-Ops Forum, from John Eden-Eadon's paper on the Sypher Suites.
"The front half of the room is dominated by the SSL5000 desk. This is a forty-two channel, computer-controlled desk made by Solid State Logic. The desk is the first to be made by SSL with motorised faders, with the standard 0-30 scale used by Television Sound."
Gary Clarke also wrote about the SSL faders: *"The quadrant faders had massive problems with the motors having to move the fader over a curve. They would frequently burn out. "*
[19] The Camden Theatre details are on a wonderful website, stuffed with information on the history of BBC Radio gear: http://www.orbem.co.uk/outsidestds/camden.htm
[20] From 'Profile: Joe Young by Adrian Hope: Studio Sound November 1973. (note the website https://jazzinbritain.org/bbc-jazz-club/ lists this session as being on 08/11/1971…two years before the Studio Sound article)

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